

A Skip Connection Generative Adversarial Network With EEG Background Noise for Motor Imagery Decoding

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Abstract—EEG decoding is a key for motor imagery-based brain–computer interfaces (MI-BCIs) to decode people’s motor intentions, holding a broad potential in medical rehabilitation. However, the amount of MI EEG data is often small because of some practical EEG collection limitations, which decreases the EEG decoding performance. This study proposes a skip connection generative adversarial network (GAN)-based data augmentation model to generate MI EEG samples, to expand the amount of the samples for the EEG decoding. The SGD utilizes independent component analysis (ICAs) to extract EEG background noise and to retain more biological information for the GAN and then employs a skip connection and multihead attention to integrate high- and low-level EEG features in the generator. Subsequently, the SGD adopts a temporal convolutional network (TCN) to discriminate between samples. The brain–computer interface (BCI) Competition IV-2a dataset is used to validate the SGD on four-class MI EEG decoding. The results show that the SGD can retain high accuracies (best accuracy: 86.15%, $p = 0.034$) under different ratios of generated MI EEG samples, implying that the skip connection and multihead attention are helpful for reconstructing the temporal and spatial features of MI EEG by ablation experiment. This study demonstrates the feasibility of applying EEG background noise to generate MI EEG samples, providing a new way to solve the small sample problem faced by the MI-BCI.

Index Terms—Brain–computer interface (BCI), EEG background noise, EEG decoding, generative adversarial networks (GANs), skip connection.

I. INTRODUCTION

BRAIN–COMPUTER interfaces (BCIs) enable direct communication between the human brain and computers by decoding neural activity [1], highlighted as one of the

seven technologies to watch in 2024 [2]. As a new type of human–machine interaction (HMI) technology, BCI has received increasing attention. It has been employed in the military, medical field, and smart HMI [3], [4], [5]. Motor imagery-based BCI (MI-BCI), a widely used BCI [6], requires the subject to imagine motor movements to produce desynchronization/synchronization (ERD/ERS) characteristics in the brain [7], [8] that are used to “decode” the subject’s motor intentions. The MI-BCIs have great prospects in medical rehabilitation and peripheral control [9], [10], while complex MI EEG features pose challenges to EEG decoding. With the rapid development of computer science, various machine learning models, especially deep learning (DL) models, have been progressively applied to MI EEG decoding, such as convolutional neural network (CNN), long short-term memory (LSTM) network, and hybrid CNNs [11], [12], [13]. Lawhern et al. [14] proposed a compact CNN, EEGNet, for EEG decoding. Li et al. [15] proposed a neural network feature fusion algorithm combining CNN and LSTM.

Although the DL has demonstrated great potential in EEG decoding, it is still faced with the challenge of a small sample problem that limits the development of MI-BCI. Small sample is a bottleneck to the EEG decoding because of some practical limitations of EEG collection, such as high-cost and time-consuming MI-BCI experiment, subject’s fatigue, and intrasubject and intersubject variability [16] of EEG data. The small number of samples makes it hard for the EEG decoding model to capture and learn inherent or important EEG features and to yield great performance.

Generative adversarial network (GAN) is a promising approach to address the small sample problem by generating some artificial MI EEG samples to expand the training dataset. Hartmann et al. [17] first applied GAN to generate EEG data for a single channel. Since then, GAN-based EEG data generation has gained more and more attention in the BCI field. First, some researchers tried to generate stimulus-related EEG data. Zhang et al. [18] constructed a Wasserstein GAN for generating high-quality ERP data, which innovatively utilized resting-state EEG data, a representative of an individual’s basic brain state, as an input instead of Gaussian noise, to ensure the authenticity of the generated ERP data. Panwar et al. [19] proposed a WGAN with gradient penalty (WGAN-GP) to generate 1-channel and 64-channel data similar to EEG signals

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acquired under the RSVP-BCI experiment. Aznan et al. [20] trained a deep convolutional GAN (DCGAN) for generating EEG data containing SSVEP EEG signals. Second, compared to the stimulus-related EEG, spontaneous EEG data generation seems to be more challenging since the feature is more complex. In emotion recognition, Luo et al. [21] constructed a conditional WGAN that generates power spectral density and differential entropy features commonly used in emotion recognition tasks from uniform noise. Zhang et al. [22] proposed a multigenerator single-discriminator conditional WGAN. In the field of MI, Xie et al. [23] constructed a GAN combining an LSTM and a CNN to achieve the modeling of MI data in both temporal and spatial dimensions. Zhang et al. [24] proposed a DCGAN for generating MI feature maps, which are composed of raw EEG data from C3, C4, and Cz channels after short-time Fourier transform in splicing, instead of generating raw MI data. Roy et al. [25] used a bidirectional LSTM unit in both the generator and discriminator of GAN. These studies sufficiently demonstrate that the GAN is effective in EEG data generation and decoding improvement.

Current studies mainly apply Gaussian or uniform noise as the input of generator of the GAN to generate the EEG data. Different distributions of the noise and the EEG data increase computational burdens to the GAN. It might become difficult to obtain generated EEG data that are similar to authentic EEG data. In addition, most studies focus on extracted EEG features generation rather than raw EEG data, which is possible to loss some inherent physiological information. The input data to the generator of the GAN are key to the quality of the generated data, so how to design the input data is important.

In fact, collected data not only contain ERD/ERS features that are correlated to MI tasks but also include EEG background noise. The EEG background noise contains continuous neural oscillations, spontaneous brain rhythms, and subtle temporal variations, which might be more suitable than the noise as the input to the GAN. Therefore, this study assumes that applying the EEG background noise to generate MI EEG data (samples) might be superior to applying the random noise in the GAN and could improve the EEG decoding further. However, when generating raw EEG data, the stability and convergence issues associated with GAN training remain a concern. The skip connection allows the decoder of the GAN to fuse the feature maps from the corresponding positions of the encoder during each level of upsampling. By providing a more direct propagation path for the gradients, the problems of gradient vanishing as well as the spatiotemporal information loss might be solved. Therefore, this study tries to construct an encoder-decoder incorporating a skip connection for dealing with EEG background noise.

Considering the small sample problem in the MI-BCI and the advantages of GAN, this study proposes a skip connection GAN-based data augmentation model to generate MI EEG samples, to expand the amount of the samples for the EEG decoding. The SGD utilizes independent component analysis (ICAs) to extract EEG background noise from the raw EEG data to retain biological information for the GAN. Then, a skip connection and multihead attention are applied to integrate high- and low-level EEG features in the generator.

Temporal convolutional networks (TCNs) are combined in the discriminator to distinguish authentic from generated data. The proposed SGD is validated on the BCI Competition IV-2a dataset with a four-class MI EEG decoding test. The proposed model can retain high accuracies (best accuracy: 86.15%, $p = 0.034$) under different ratios of generated MI EEG samples to total samples. The main contributions of this study are as follows.

- 1) This study innovatively utilizes EEG background noise as the input of the GAN to generate raw MI EEG data. It extracts the background noise by ICA to retain subject's biological information, making the generated data more similar to the authentic EEG data. Using the EEG background noise is demonstrated to improve the MI EEG decoding accuracy.
- 2) An architecture of combining the skip connection and a multihead attention mechanism is proposed when constructing the encoder and decoder of the generator in the SGD model, which ensures that both low-level details and high-level global features of the EEG are generated. The ERD/ERS characteristics extracted from the generated data indicate that this architecture is helpful to retain the important EEG biomarkers of MI brain activity. This study demonstrates that applying the EEG background noise as an input of GAN to generate raw MI EEG data is feasible, providing a new way to solve the small sample problem faced by the MI-BCI.

The organization of this study is as follows. Section II describes the architecture and every module of the proposed SGD. Section III introduces the validation experiment of the SGD on the aspects of public dataset introduction, validation procedure, and evaluation metrics. Section IV shows the EEG decoding performance with different ratios of generated data and ablation experiment results. Section V mainly discusses the comparison of GAN research and the limitation and future work. Section VI is the conclusion of this study.

II. METHODS

This study proposes an SGD model for MI EEG decoding (Fig. 1) that comprises three modules: an EEG background noise (EBN) extraction module, the SGD, and a multibranch multiattention CNN (MBMANet). First, the EBN utilizes ICAs to extract EEG background noise from original EEG data that serve as the input for the subsequent SGD. Then, the SGD takes the noise as input to accomplish adversarial training between the skip connection generator and the temporal convolution discriminator. The generator transforms the noise into generated EEG samples, aiming to expand the amount of MI EEG samples. Finally, the MBMANet is trained using both authentic and generated EEG samples to yield an MI EEG decoding model. The generated data would aid the decoding model in better decoding the subject's four motor intentions for the MI-BCI.

A. EEG Background Noise Extraction Module

The EEG background noise is extracted by the EBN extraction module using ICA technology. Prior to conducting

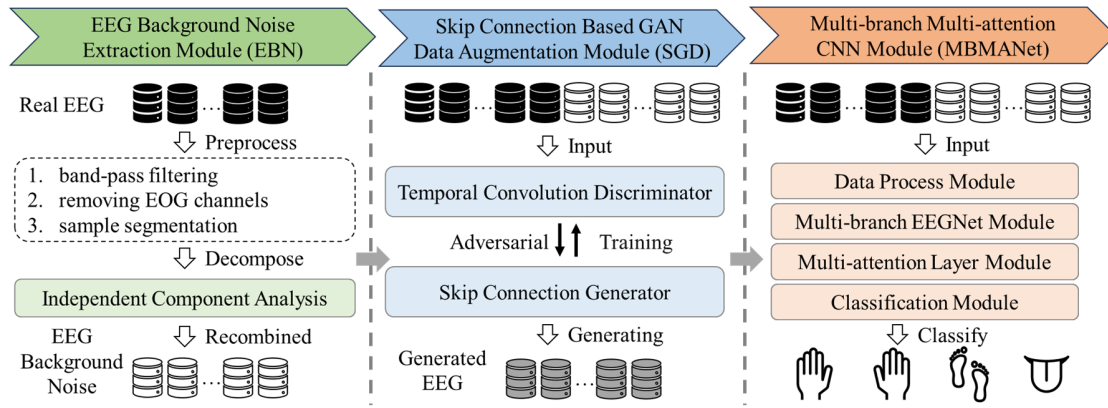


Fig. 1. Framework of EEG decoding in an MI-BCI. The decoding includes three stages: EBN (used for extracting EEG background noise), SGD (used for generating MI EEG data), and MBMANet (used for machine learning and EEG decoding).

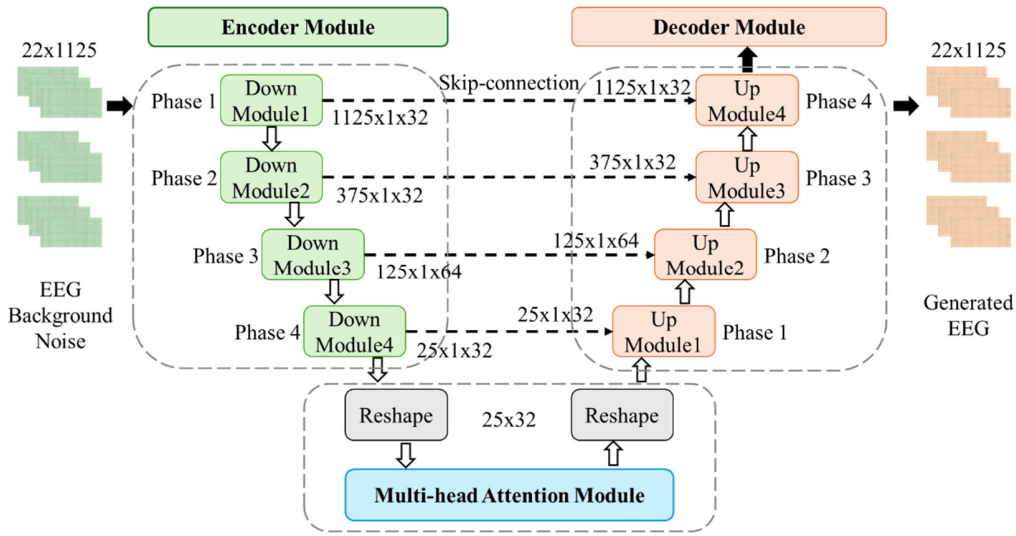


Fig. 2. Generator network based on skip connection.

ICA on the subjects' MI EEG signals, preprocessing steps are undertaken, including bandpass filtering between 8 and 30 Hz, the removal of channels related to electro-oculogram (EOG), and segmentation of MI EEG data. Then, the EBN modules conduct ICA that regards the collected brain signal as a linear combination of several independent source signals (components).

The EBN first employs the FastICA [26] to decompose collected MI EEG data into multiple components, such as MI-related components, electromyographic interference, EOG artifacts, and electrical circuit noise. The number of independent components (ICs) is set to the number of EEG channels. Then, the weights of MI-related components and noise are adjusted. Among them, the MI task-related ICs are given high weights to ensure that the generator learns the basic neural patterns related to MI. The ICs, such as EOG and EMG artifacts, as well as environmental noise, are given small weights for increasing the actual variability of the input data, preventing the generator from simply copying the inputs, and facilitating its learning of effective mapping relations. The combination of these weight-adjusted ICs is the EEG background noise. Finally, the EBN module provides the

EEG background noise as input data for the generator in the SGD module.

B. Skip Connection Based on GAN Data Augmentation Module

The SGD is the core module to generate MI EEG data that includes a generator and a discriminator. The generator employs skip connection (skip connection generator) to pass low-level details and local feature information to corresponding expansion paths, to learn more biological information. The discriminator uses a temporal convolution module (temporal convolution discriminator) to capture temporal dependencies in EEG data, effectively distinguishing subtle differences between generated and authentic EEG data to enhance authenticity. By the encoder and discriminator, the GAN generates the amount of EEG data.

1) *Skip Connection Generator*: The skip connection generator network is a core component of the SGD, tasked with transforming EEG background noise into MI EEG data. The design of this skip connection generator, as shown in Fig. 2, is inspired by the U-Net architecture proposed by

TABLE I
ENCODER NETWORKS IN THE SKIP CONNECTION GENERATOR

Module	Layer type	# filter	size	Output	Module	Layer type	# filter	size	Output
Module 1	Input	-	-	(22, 1125)	Module 3	Conv2D	64	(16, 1)	(375, 1, 64)
	Reshape	-	-	(1125, 22, 1)		BatchNorm	-	-	(375, 1, 64)
	Conv2D	16	(64, 1)	(1125, 22, 16)		Activation	-	-	(375, 1, 64)
	BatchNorm	-	-	(1125, 22, 16)		AveragePool2D	-	(3, 1)	(125, 1, 64)
	DepthwiseCon2D	32	(1, 22)	(1125, 1, 32)		Dropout*	-	-	(125, 1, 64)
	BatchNorm	-	-	(1125, 1, 32)		Conv2D	32	(4, 1)	(125, 1, 32)
	Activation	-	-	(1125, 1, 32)		BatchNorm	-	-	(125, 1, 32)
	Conv2D	32	(64, 1)	(1125, 1, 32)	Module 4	Activation	-	-	(125, 1, 32)
Module 2	BatchNorm	-	-	(1125, 1, 32)		AveragePool2D	-	(5, 1)	(25, 1, 32)
	Activation	-	-	(1125, 1, 32)		Dropout*	-	-	(25, 1, 32)
	AveragePool2D	-	(3, 1)	(375, 1, 32)		Reshape	-	-	(25, 32)
	Dropout*	-	-	(375, 1, 32)		/	-	-	

Ronneberger et al. [27] and is structured into an encoder, a multihead attention module, and a decoder.

a) *Encoder*: The encoder utilizes a series of convolutional and pooling layers to perform deep feature extraction on the input EEG background noise, progressively reducing the spatial dimensions of the data to lay the foundation for generating authentic and diverse MI EEG data. The entire encoder consists of four downsampling modules. In Module 1, the input data are first reshaped into 3-D data, followed by the application of 16 convolutional of size (16, 1) and 32 convolutional kernels of size (1, 22) for 2-D convolution and depthwise separable convolution, respectively. The operations in Modules 2–4 are the same, involving feature extraction using 2-D convolutional kernels and dimensionality reduction of feature maps through average pooling layers. The difference lies in the number and size of convolutional kernels used in each module. All modules in the encoder utilize the ELU activation function, with batch normalization layers added after 2-D convolutional layers to accelerate model training. Finally, the feature map output by the encoder is transformed into 2-D data by a reshape layer, which then serves as the input for the multihead attention mechanism layer. Table I provides a detailed overview of the network layers used in each module of the encoder, the sizes and quantities of convolutional kernels, and the output shapes of each layer.

b) *Multihead attention module*: By parallel learning of different subsets of features extracted by the decoder, the method meticulously captures the essential differences between EEG background noise and authentic MI EEG characteristics. Through adaptive weight adjustments, it emphasizes key features that enhance the authenticity of the generated MI EEG data while disregarding less important background noise information. This approach helps to mitigate the issue of homogenization among different generated samples.

c) *Decoder*: Utilizing the EEG background information features extracted by the encoder, the decoder reconstructs the feature information of authentic EEG data through a series of upsampling and transposed convolution steps to generate authentic MI EEG samples. The decoder is the

inverse structure of the encoder, consisting of four upsampling modules. Each module includes transposed convolution layers that gradually increase the spatial dimensions of the data, batch normalization layers that stabilize the learning process, and ELU activation functions that introduce nonlinear characteristics. The parameters used in each module of the decoder, such as the number and size of convolution kernels, vary. Skip connection plays a crucial role throughout the generator by transferring features from each module of the encoder to the corresponding upsampling module in the decoder. This allows for the comparison of features between different levels of original EEG background noise input and reconstructed data. Based on feedback from the discriminator network, this configuration better isolates irrelevant noise and retains key features of the authentic MI EEG data.

In summary, the skip connection generator utilizes an encoder, multihead attention, decoder, and skip connection to generate MI EEG samples.

2) *Temporal Convolution Discriminator*: The discriminator based on TCNs is a crucial component of the SGD, tasked with distinguishing whether the input data are authentic or generated samples to guide the generator in synthesizing MI EEG data. The discriminator, shown in Fig. 3, consists of three parts: convolution block (CB) module, TCN module, and output block (OB) module.

a) *CB module*: The initial module of the discriminator network follows the architecture of the EEGNet model proposed by Lawhern et al. [14]. By utilizing three different types of convolutional layers, it effectively encodes low-level spatiotemporal information within MI EEG data while reducing the model's complexity and number of parameters. Finally, it outputs feature maps with advanced temporal representations to serve as an input for the TCN module.

b) *TCN module*: This is the core module of the discriminator network and captures long-range temporal dependencies by stacking multiple residual blocks. It learns crucial ERD/ERS dynamic changes within EEG data, providing distinctive feature inputs for the subsequent output module to differentiate between authentic and generated samples.

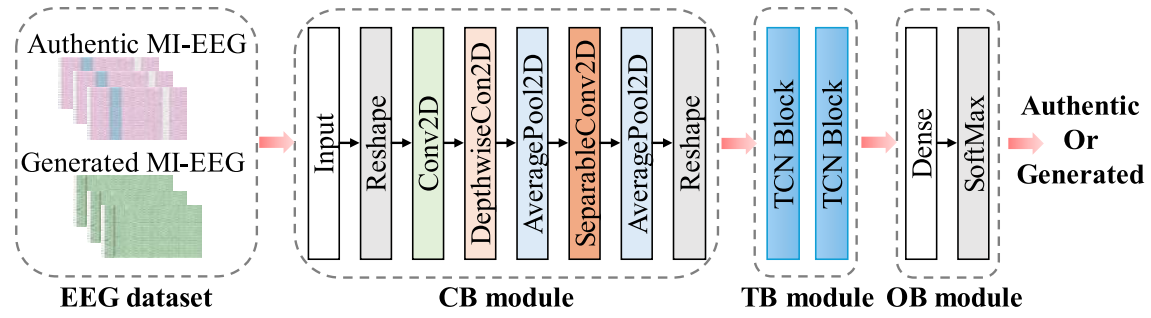


Fig. 3. Temporal convolutional discriminator.

TABLE II
NETWORK STRUCTURE OF TEMPORAL CONVOLUTIONAL DISCRIMINATOR

Module	Layer type	# filter	size	Output	Module	Layer type	# filter	size	Output
Conv Block	Input	-	-	(C, T)	TCN Block (*L)	Conv1D	F_T	K_T	$(T // 64, F_T)$
	Reshape	-	-	$(T, C, 1)$		BatchNorm	-	-	$(T // 64, F_T)$
	Conv2D	F_1	$(K_E, 1)$	(T, C, F_1)		Activation	-	-	$(T // 64, F_T)$
	BatchNorm	-	-	(T, C, F_1)		Dropout*	-	-	$(T // 64, F_T)$
	DepthwiseCon2D	$D * F_1$	$(1, C)$	$(T, 1, D * F_1)$	Output Block	Dense	-	-	N
	BatchNorm	-	-	$(T, 1, D * F_1)$		SoftMax	-	-	N
	Activation	-	-	$(T, 1, D * F_1)$					
	AveragePool2D	-	$(8, 1)$	$(T // 8, 1, D * F_1)$					
	Dropout*	-	-	$(T // 8, 1, D * F_1)$					
	SeparableConv2D	F_2	$(16, 1)$	$(T // 8, 1, F_2)$					
	BatchNorm	-	-	$(T // 8, 1, F_2)$					
	Activation	-	-	$(T // 8, 1, F_2)$					
	AveragePool2D	-	$(8, 1)$	$(T // 64, 1, F_2)$					
	Dropout*	-	-	$(T // 64, 1, F_2)$					
	Reshape	-	-	$(T // 64, F_2)$					

c) *OB module*: This module is composed of a fully connected layer and a SoftMax layer. The former one is to integrate the features extracted by the TCN module, and the latter one outputs the final discrimination result, determining whether the input data are an authentic or a generated sample.

Table II provides a detailed description of the structure of the temporal convolution discriminator network, including the network layers used in each module, the number and size of convolution kernels, and the output shapes. $C = 22$ and $T = 1125$ represent the number of channels and sample points in the EEG samples, respectively. F_1 , F_2 , and D correspond to the number of temporal filters, pointwise filters, and the depth multiplier, respectively. K_E is the size of the first convolution kernel in the CB module; K_T and F_T are the size and number of convolution kernels in the temporal convolution network module, respectively; and $L = 2$ is the number of layers in the TCN within the temporal convolution network module. The dropout rates in the convolution and temporal convolution modules are set at 0.3 and 0.25, respectively, and the activation function used is ELU. $N = 2$ represents the number of classification categories, with the SoftMax function outputting

the probability that a sample belongs to either the authentic or generated category.

C. Multibranch Multiattention CNN

This article employs an MBMANet as an EEG decoding model (Fig. 4) that exhibits good performance on MI EEG decoding in our previous study [28]. It consists of four main modules: data preprocessing module, multibranch convolutional network module, multiattention mechanism module, and classification module. The data preprocessing module serves as an input module of the MBMANet, taking raw MI EEG data as input and extracting EEG data from the time window of interest. It normalizes the data on each channel based on the mean and standard deviation to eliminate amplitude discrepancies in the raw data; the multibranch convolutional network module is the feature extraction module of the MBMANet, where each branch receives the same EEG data and extracts EEG features in parallel through three EEGNet branches with different convolutional kernel sizes or numbers, outputting diverse feature maps to the multiattention mechanism module; the multiattention mechanism module, serving as the core

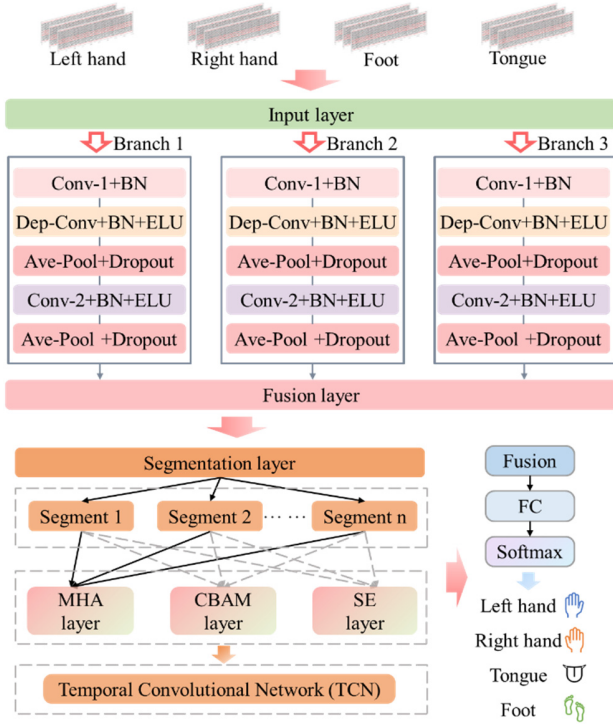


Fig. 4. Proposed MBMANet model.

of the MBMANet, receives output from the multibranch convolutional network module. It uses a segmentation layer to divide the input features and introduces various attention mechanisms for weight adjustment, along with TCNs to extract advanced temporal features. The classification module is the output module of the MBMANet. It integrates the advanced temporal features extracted from different branches via a fully connected layer and then utilizes a SoftMax layer to output the final four-class classification result.

III. EXPERIMENT

A. Dataset

The BCI Competition IV dataset 2a (BCI IV 2a) is a publicly available multiclass MI dataset that contains EEG recordings derived from nine subjects engaged in imagined movements of the left hand, right hand, feet, and tongue. Fig. 5 depicts the temporal schedule of the BCI IV 2a dataset. Data were collected over two separate sessions for each subject using 22 EEG electrodes at a sampling frequency of 250 Hz, with each session comprising 288 trials evenly distributed across each class. The data acquisition protocol for each trial is as follows. At $t = 0$ s, a fixed cross is displayed on the black screen of the computer, accompanied by a short auditory beep signaling the commencement of the trial. At $t = 2$ s, an arrow is displayed on the screen to indicate the required movement. Left, right, up, and down correspond to the movements of left-hand movement, right-hand movement, feet movement, and tongue movement. The arrow remains on the screen for 1.25 s, during which the subject performs the MI task for a duration of 4 s. The trial ends when the fixed cross on the screen disappears ($t = 6$ s).

In this study, C , T , and N_c in each trial $X \in \mathbb{R}^{C \times T}$ and its corresponding label $y \in \{1, 2, \dots, N_c\}$ are set to 22, 1125, and 4, respectively, representing 22 EEG electrode channels, 1125 sample points (EEG segment data of 4.5 s from the previous 0.5 s of cue to the end of trial in a trial), and four MI task classes (classes 1–4 correspond to the left hand, right hand, foot, and tongue, respectively).

B. Training and Validation Strategies

During SGD module training, a dual time-scale updating rule is employed. The learning rates of the generator and discriminator are set to 0.001 and 0.0002, respectively. The alternating training ratio is set to 3:1, and Adam is used as the optimizer with a batch size of 72.

This study designs a three-stage training strategy for enhancing the diversity of generated EEG data from the SGD module. The target subject is defined as the subject who needs to generate EEG data, and the other subjects are nontarget subjects.

- 1) *Stage 1*: In the initial training stage, the SGD is trained only using MI EEG data from the target subject. The core purpose of this stage is to allow the SGD to focus on learning the typical EEG features of the target subject.
- 2) *Stage 2*: The nontarget subject's EEG data are first processed for artifact and noise removal, and their amplitudes are randomly scaled to make them closer to the target subject's data. Then, these scaled data are stacked with the target subject's EEG data at preset weights to expand the target subject's dataset. The main goal of this stage is to prevent SGD from overfitting the target subjects' MI data by introducing a more diverse training dataset.
- 3) *Stage 3*: The training set of the SGD is once again limited to MI EEG data from the target subject. The training in this stage aims to fine-tune the model further and optimize its fit with the target subject's data. The distribution of MI EEG data at different training stages is shown in Table III.

This study also designs a validation strategy that adds different ratios of generated EEG data to validate the effectiveness of the SGD module. For each subject, the authentic data used to train the MBMANet module are 288, and the generated data used for training involve nine different ratios, that is, no addition (generated data = 0), the addition of 72, 144, 216, 288, 360, 432, 504, and 576 generated data, corresponding to the ratio group as a baseline, 1:0.25 (authentic data: generated data), 1:0.5, 1:0.75, 1:1, 1:1.25, 1:1.5, 1:1.75, and 1:2, respectively.

All models are implemented in Tensorflow 2.7 by using an Intel Xeon Platinum 8255 C CPU (2.50 GHz) and an NVIDIA RTX 3080 GPU with 10-GB RAM.

C. Evaluation Metrics

This study employs classification accuracy (ACC), receiver operating characteristic curve (ROC), and area under the curve (AUC) for assessment.

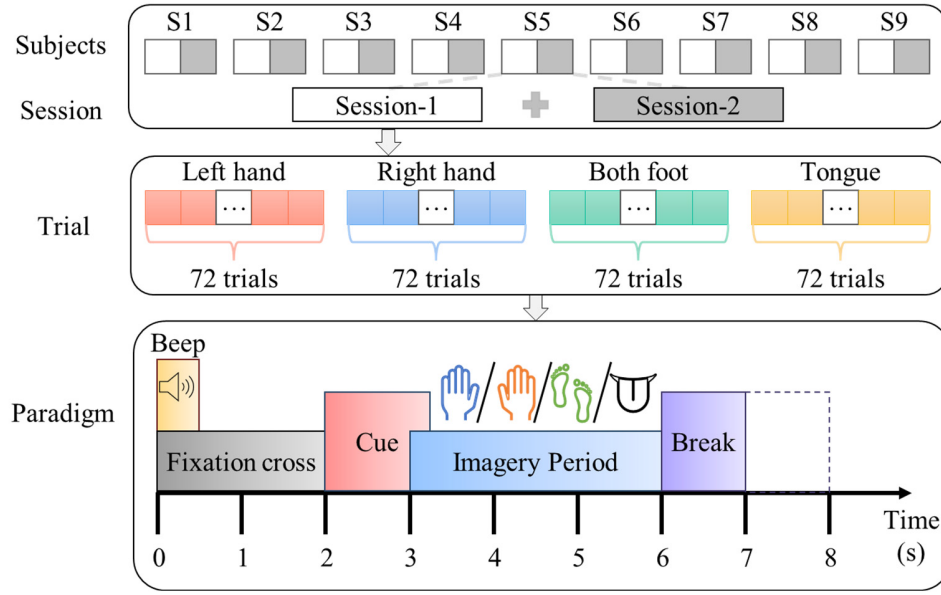


Fig. 5. Timing scheme of the BCI IV 2a.

TABLE III
DATA DISTRIBUTION FOR DIFFERENT TRAINING STAGES OF SGD

Stage	Description	Subject	Number of MI EEG Samples
1	Focused learning on target subject data	1 subject	288 samples
2	Data augmentation and model generalization	9 subjects	2592 samples
3	Refocused learning on target subject data	1 subject	288 samples

1) *ACC*: It is the most widely used metric in MI-EEG decoding and is defined as

$$\text{Acc} = \frac{1}{N_c} \times \sum_{i=1}^{N_c} (\text{TP}_i / S_i) \quad (1)$$

where TP_i is the number of correctly predicted samples in class i , S_i is the number of samples in class i , and $N_c = 4$ indicates the number of classes.

2) *ROC*: It is used to visualize the decoding model's ability to classify four types of MI-EEG samples at different thresholds, as well as the sensitivity and specificity of the decoding model. The ROC is determined by two parameters: true positive rate (TPR) and false positive rate (FPR). The calculations for TPR and FPR are as follows:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (2)$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (3)$$

3) *AUC*: The area under the ROC curve is used to quantify the ability of the decoding model to distinguish between positive and negative samples, with a value ranging from 0 to 1. The closer the value is to 1, the better the classification performance of the decoding model.

4) *Statistical Test*: To reliably evaluate the performance of the EEG encoding models, the Shapiro-Wilk test assesses the normality of the distribution of classification accuracies

(ACC) for the decoding models, with a p-value greater than 0.05 indicating that the normality assumption is satisfied. Mauchly's test of sphericity is utilized to assess the assumption of sphericity required. If the data meet the prerequisites of normality and sphericity, a one-way repeated measures analysis of variance (ANOVA) is conducted to compare the performance differences among the different groups. The Bonferroni correction is applied to control the family-wise error rate and maintain the statistical significance level when performing multiple comparisons. If the normality assumption is not satisfied, the Friedman and Wilcoxon signed-rank tests with Bonferroni correction are used for nonparametric statistical analysis. The significance level is set at 0.05 to determine the statistical significance of the analyses.

5) *ERD/ERS Evaluation*: Since the ERD/ERS is an important EEG characteristic of MI brain activity and is always extracted to decode the subject's intention, this study uses it to evaluate whether the generated data own this biological information. It is derived from data from channels C3, Cz, and C4, and the time window is $[-2, 4.5]$ s of the cue [29]. It is calculated as the percentage decrease or increase in power relative to a reference period as follows:

$$\text{ERD/ERS} = \frac{A - R}{R} \quad (4)$$

where the power during the post-event period within the frequency band of interest is denoted by A and the power during the reference period is denoted by R .

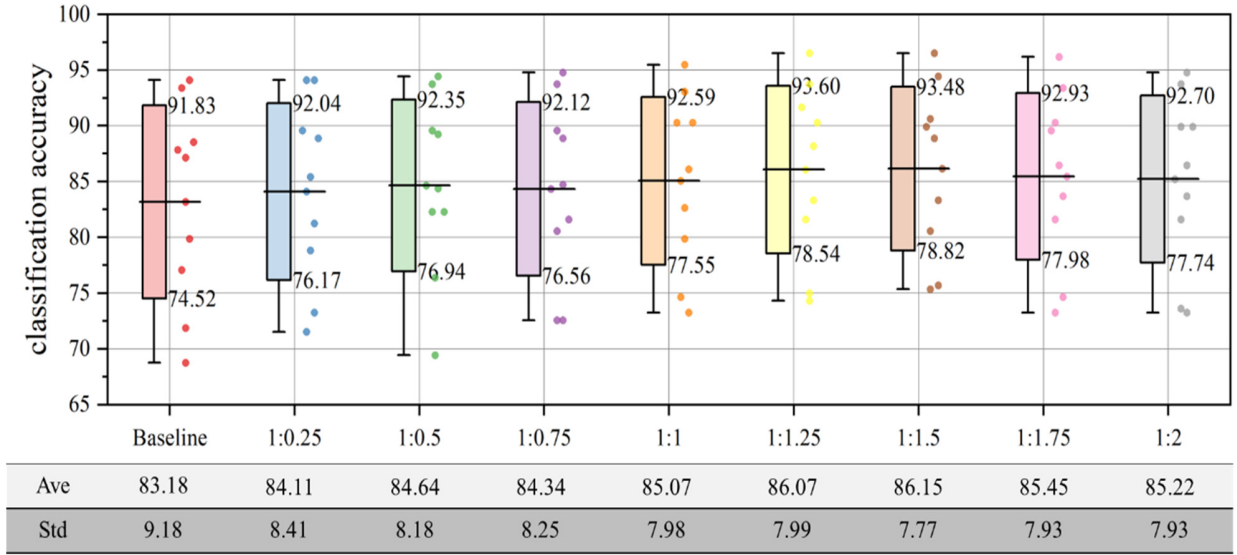


Fig. 6. Boxplot of the ACC after adding different ratios of generated data. The x -axis lists nine groups with different ratios (defined as authentic data to generated data). The ratio of “Baseline” is 1:0.

IV. RESULT

A. EEG Decoding Accuracy With Different Ratios of Generated Data

To validate whether the ratio of the generated data influences the MI EEG decoding performance, this study sets up nine groups with varying ratios of generated data. Fig. 6 lists their corresponding ACCs. The boxplot displays the range of the box, median, and whiskers corresponding to the standard values, mean values, and minimum–maximum values, respectively. Overall, all different ratio groups of adding generated data achieve improvements, higher than the baseline. This indicates that using MI EEG samples generated by SGD for data augmentation can enhance the ACC of the decoding model.

Comparing different ratio groups, the 1:1.5 ratio group exhibits the highest average ACC of 86.15%, whereas the baseline obtains the lowest average accuracy of 83.18%. Compared to the baseline group, the accuracy steadily increases in the 1:0.25, 1:0.5, and 1:0.75 ratio groups, reaching the highest ACC in the 1:1.5 ratio group with a 2.93% improvement.

However, as more generated data are added, the average ACC tends to decrease. In addition, the average ACC of the 1:0.75 ratio group is lower than that of the 1:0.5 group, indicating that a larger ratio of generated samples would degrade the learning ability of the decoding model. This is likely due to the fact that the quality of the generated data is not as well as the authentic data. Therefore, it is necessary to limit the ratio of the generated data in the training set.

B. Generality of the SGD Model

Due to individual differences in MI EEG, this study further investigates whether this proposed SGD model is generalized for each subject. Fig. 7 lists each subject’s decoding accuracy under different ratios. The subjects’ ACC does increase as generated data is added, and AUC values and microaverage

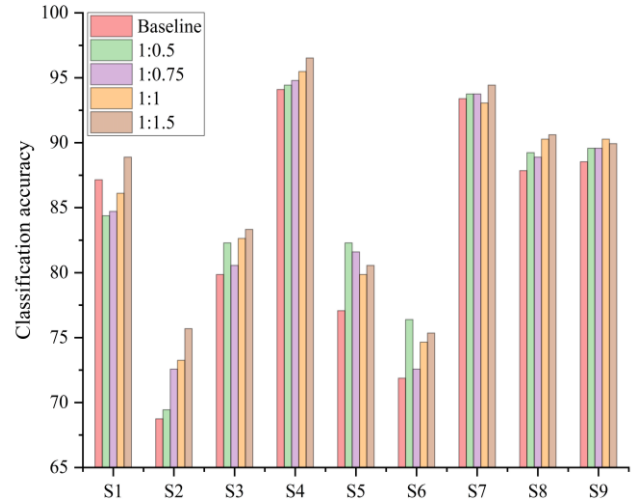


Fig. 7. Comparison of ACC after adding different ratios of generated data in different subjects.

ROC curves for all subjects could be above 0.90 in Fig. 8, demonstrating the effectiveness of the SGD model.

The individual difference is also observed in our results. The ACC does not consistently increase with the addition of generated data. Compared to the 1:0.75 ratio, most subjects show improved ACC in the 1:1 ratio but subjects S5’s and S7’s decrease. Except for S9, all subjects achieve the highest ACC in the 1:1.5 ratio. S1, S3, and S7 exhibit high ACC across almost all MI movements, with all ROC curves tending toward the upper left corner. Particularly for S3, perfect ACC is achieved for MI EEG data related to right hand, tongue, and foot, with corresponding AUC values of 1.0. This indicates that SGD accurately learns and reconstructs the discriminative features of MI EEG samples for S3’s right hand, foot, and tongue, effectively enhancing the diversity of training set and boosting the generalization ability of the decoding model. In contrast, S5 demonstrates poorer performance for foot and

TABLE IV
STATISTICAL ANALYSES OF DIFFERENT RATIO GROUPS ON ACC

Tests	baseline	1:0.25	1:0.5	1:0.75	1:1	1:1.25	1:1.5	1:1.75	1:2
Shapiro-Wilk	0.410	0.448	0.609	0.419	0.561	0.508	0.452	0.686	0.343
Levene	0.998 (All ratio group)								
Mauchly	0.616 (All ratio group)								

Note: The Shapiro-Wilk test assesses the normality within each individual ratio group, whereas the Levene test and the Mauchly test evaluate the homogeneity of variances and the sphericity assumption, respectively, across all ratio groups collectively.

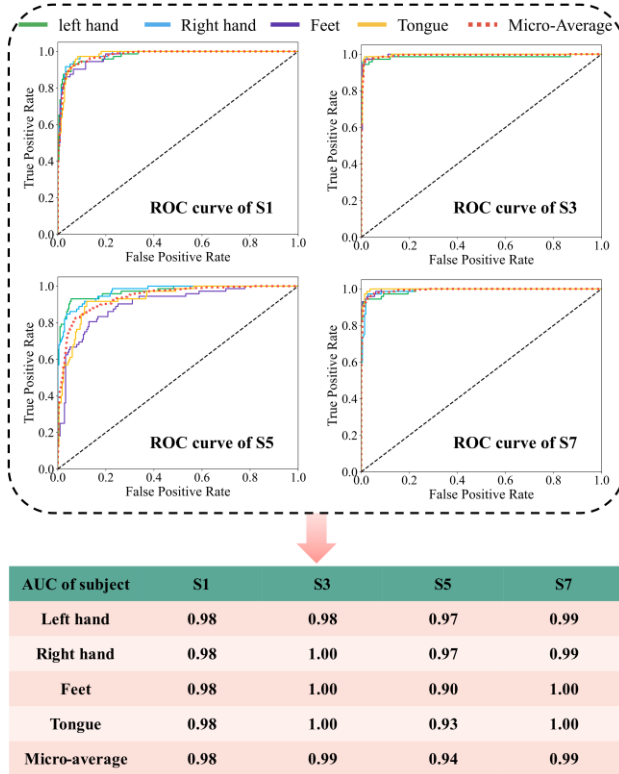


Fig. 8. ROC curves and AUC values for S1, S3, S5, and S7 subjects in the 1:1.5 ratio group. The upper part of the figure displays the ROC curves for left hand, right hand, foot, tongue, and microaverage, while the lower part lists the corresponding AUC values.

tongue, with AUC values of only 0.90 and 0.93, respectively, lower than the other three subjects. This could be attributed to less prominent EEG features of the foot and tongue for subject S5, making it challenging for the decoding model to distinguish.

Although the SGD improves the decoding accuracy for all the subjects and has demonstrated its generality, it is worth noticing the individual difference of MI EEG data. This is because each subject has his/her own “strong” movement, which could produce better EEG data; the diversity and quality of the EEG data of the four classes are different. It is necessary to set a subject-specific ratio for each subject when adding the generated data.

C. Statistical Analysis

To further evaluate whether adding different ratios of generated data significantly influences the performance of the

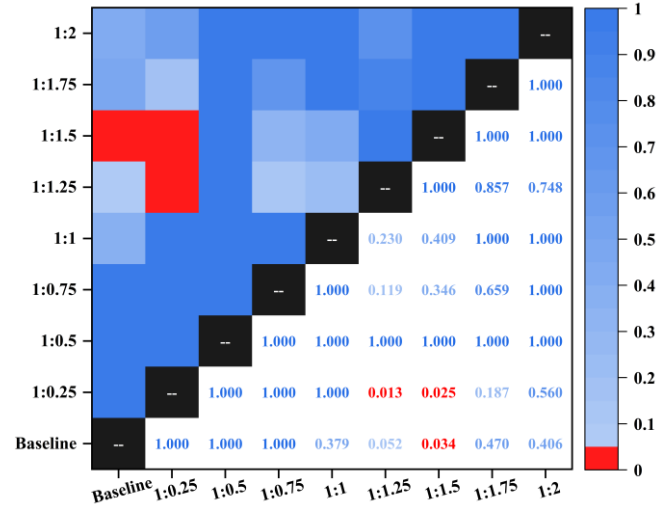


Fig. 9. Bonferroni-based corrected p-values between groups of different ratios. The upper triangular matrix shows color-coded p-values, while the lower triangular matrix displays the exact numerical p-values for pairwise comparisons. The red cell indicates significant differences ($p < 0.05$), while the blue cell indicates nonsignificant differences.

decoding model, statistical tests are employed to calculate p-values on ACC. The Shapiro-Wilk test, Levene’s test, and Mauchly’s test of sphericity confirmed that all ratio groups’ ACCs met the assumptions of normality ($p > 0.05$), homogeneity of variances ($p = 0.998$), and sphericity ($p = 0.616$), respectively, as shown in Table IV. Consequently, a one-way repeated measures ANOVA is conducted to compare performance differences among the different ratio groups. The within-subject effects ($\chi^2 = 66.161$, $df = 8$, $F = 6.775$, and $P < 0.001$) indicate that different ratios of generated data significantly impact the ACC.

Bonferroni correction is applied during post hoc pairwise comparisons to control the family-wise error rate and identify significant differences between ratio groups. Fig. 9 presents a heatmap of Bonferroni-corrected p-values for pairwise comparisons between groups with different ratios. First, most pairwise comparisons yield very high p-values (close to 1.000). For instance, the pairwise comparisons involving the 1:0.5 ratio group with other groups all have p-values of 1.000; the comparisons of the 1:1 ratio group with other groups also have p-values more significant than 0.2, far exceeding 0.05. Second, from the pairwise comparisons between the baseline group and other ratio groups, most groups with added generated data did not significantly improve the performance of the

TABLE V
SCHEME OF THE ABLATION EXPERIMENT WITH CORRESPONDING DECODING RESULTS

Scheme	Encoder module	Decoder module	Skip connection	Multi-head Attention	Random noise	EEG background noise	Parameters	ACC	Gain
1	√	√	×	×	×	√	265,326	81.94%	-1.24%
2	√	√	×	√	×	√	267,518	82.64%	-0.54%
3	√	√	√	×	×	√	265,326	85.19%	+2.01%
4	√	√	√	√	√	×	2,839,326	65.05%	-18.13%
5	√	√	√	√	×	√	267,518	86.15%	+2.97%

MBMANet. For example, the 1:0.25, 1:0.5, and 1:0.75 ratio groups all have p-values of 1.000 compared to the baseline. Only the 1:1.5 ratio group shows a significant difference ($p = 0.034$). This demonstrates that adding generated data can improve the decoding performance of MBMANet, and the ratio of adding generated data significantly affects the improvement.

In summary, this article utilizes ACC as the core evaluation metric to explore the impact of different ratios of generated data on the performance of the decoding model. Statistical tests based on one-way repeated measures ANOVA demonstrate that the ratio of added generated data significantly affects the performance of the decoding model. Further post hoc comparisons using Bonferroni correction reveal that only a specific ratio (1:1.5) of added generated data significantly improves decoding performance.

D. Ablation Experiment

This study designs ablation experiments to verify the effectiveness of different modules. Table V shows the ACCs of different schemes with/without skip connection, multihead attention mechanism, and noise (random or EEG background noise). The evaluation metric used is the average ACC of nine subjects before and after adding 432 generated data samples. Scheme 5 represents the proposed method in this study. Gain is the ACC increase from the baseline (1:0, corresponding ACC = 83.18%) to the scheme.

Scheme 2, which does not have skip connections, has an ACC decrease of 0.54%. This suggests that adding skip connections between the encoder and decoder is effective, as it allows features from the encoder layers to be passed directly to the corresponding layers of the decoder. This allows the SGD to perceive features at different scales and retain as much physiological information as possible from the EEG background noise, which helps to generate EEG data containing authentic MI features. Scheme 3, which does not incorporate the multihead attention mechanism, shows a 0.96% decrease in classification accuracy compared to scheme 5. This demonstrates that the multihead attention mechanism is an effective module. The attention mechanism helps the SGD recognize and utilize features related to MI while ignoring irrelevant or distracting information. Moreover, a comparison of schemes 1, 2, 3, and 5 reveals that the combination of skip connections and the multihead attention mechanism yields better performance, with an improvement of 2.97%. When

random noise is used as input in scheme 4, the ACC of MBMANet decreases by 18.13%. In contrast, in scheme 5, using EEG background noise as input results in a 2.97% increase in ACC. This indicates that the EBN module is an effective component, capable of extracting biologically meaningful EEG background noise as input, thereby ensuring the quality of the generated data by the SGD and improving the performance of the MBMANet. In addition, although schemes 1, 2, 3, and 5 apply the similar number of parameters, scheme 5 achieves higher accuracy, which demonstrates that the GAN architecture is effective. Scheme 4 adopts many more parameters, but the accuracy declines, which supports our assumption that the EEG background noise is a meaningful input and could provide more inherent information to learn.

In summary, the results of the ablation experiments demonstrate the effectiveness of the skip connection, the multihead attention mechanism, and the EEG background noise added in scheme 5. These modules improve the quality of the generated EEG data and help to improve the performance of the decoding model.

V. DISCUSSION

A. Comparison With Other SOTA Models

Some researchers also apply GAN to generate MI EEG data to improve the decoding accuracy. Here, we list some other state-of-the-art (SOTA) models of GAN and focus on the comparison of model architecture, type of noise for the GAN input, and decoding accuracy, as shown in Table VI.

Since the researchers focus on different aspects of the EEG data, the architectures of the four data augmentation models are different. Song et al. [30] proposed a conditional GAN (CGAN) to identify key features by integrating SE attention mechanisms in the discriminator. Similarly, Xie et al. [23] proposed an LSTM-GAN that combined CNN and LSTM in the discriminator and added a spatial attention layer in the generator to focus mainly on spatial and temporal EEG features. Choo et al. [31] constructed a WCGAN-GP based on CGAN by using the Wasserstein distance and gradient penalty to avoid gradient vanishing during training. Instead of random noise, our SGD uses ICAs to extract EEG background noise as input to the generator. The generator integrates multilevel features of the EEG background noise using skip connection and multihead attention mechanisms. In the discriminator, a TCN module capable of capturing EEG long-time dependencies is introduced.

TABLE VI
COMPARISON OF SKIP CONNECTION GAN WITH OTHER SOTA MODELS

Work	Data Augmentation	Decoding model	Noise type	Dataset	ACC
Ours	skip connection GAN	MBMANet	EEG background	BCI IV 2a	86.15%
Song <i>et al.</i> [30]	CGAN	EEGNet	Random noise	BCI IV 2a	78.70%
Xie <i>et al.</i> [23]	LSTM-GAN	MoCNN	Random noise	BCI IV 2a	83.99%
Choo <i>et al.</i> [31]	CWGAN-GP	BiLSTM	Random noise	BCI IV 2a	83.80%

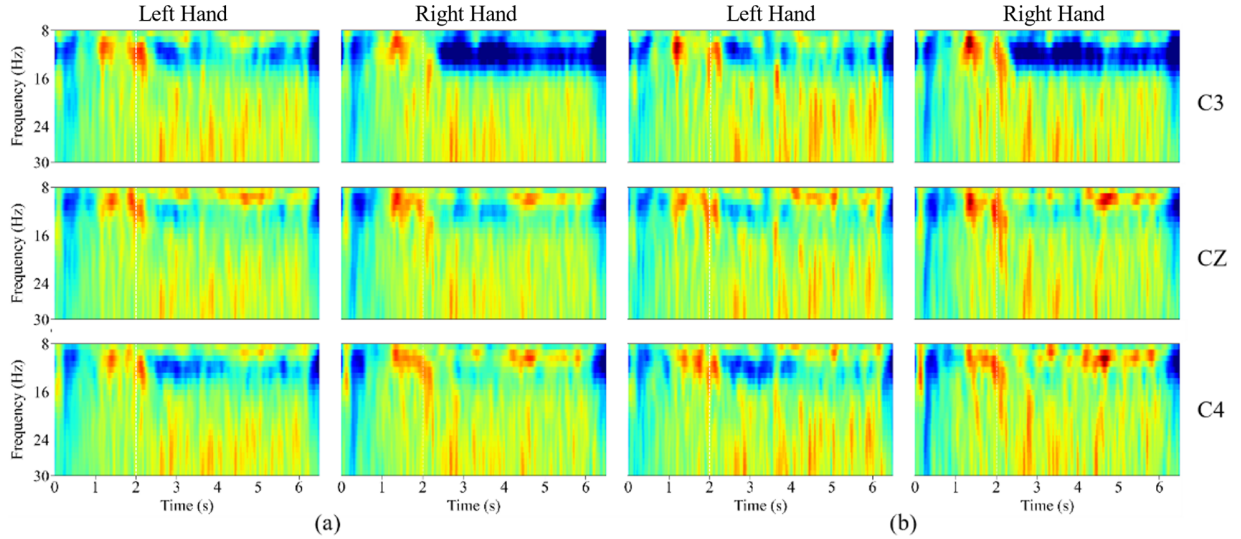


Fig. 10. Time–frequency feature maps of (a) authentic and (b) generated MI EEG samples. The x -axis in each subplot represents time (post-fixed cross 6.5), while the y -axis represents the frequency range (8–30 Hz). The blue color stands for ERD (power decrease), and the red color stands for ERS (power increase).

From the accuracy comparison, the proposed SGD shows higher EEG decoding performance on the four-class validation test of the BCI IV 2a dataset. The reasons for this superiority are possibly related to the novelties of this study. First, the SGD model uses EEG background noise as the generator input rather than random noise. The EEG background noise is extracted from authentic EEG data by ICA and contains information related to background brain activity and physiological artifacts during motor imagery, which has a higher intrinsic correlation with subject's biological information, facilitating the generation of MI EEG samples that are similar to the authentic EEG data. Second, this study carefully designs an encoder–decoder generator specifically for the EEG background noise by combining skip connection and multihead attention mechanism. The skip connection directly transmits low-level details and local features from the encoder to the corresponding expansion path in the decoder, ensuring the retention of detailed information from EEG background noise during encoding and decoding. The high-level features extracted by the encoder are fed into the decoder, enhancing the model's ability to focus on and recognize meaningful patterns of brain activity. Third, multihead attention is employed in the generator of the GAN to enhance its focus on critical features in EEG background noise, which is different from the SE and spatial attention layers used in [23] and [31]. The SE module enhances feature representations by adaptively recal-

ibrating channel weights, while multihead attention enhances feature representations by capturing complex spatiotemporal relationships of the EEG via multiple attention heads in parallel. The comparison implies that the SGD is a promising method to improve the MI EEG decoding accuracy.

B. ERD/ERS Characteristics of MI EEG Data

ERD/ERS is the main characteristic of EEG in the MI brain activity. In other words, it can be regarded as a biomarker of the MI EEG data. Whether this characteristic is obvious directly influences the performance of MI EEG decoding. Fig. 10 plots the time–frequency spectrograms of EEG data to demonstrate whether the generated data own this ERD/ERS. Subject 3's C3, C4, and Cz channels are plotted using either authentic (a) or generated (b) data.

The generated MI EEG data exhibit similar characteristics to the authentic MI EEG data as they both exhibit ERD/ERS phenomena. For the left hand during 3–6 s, the ipsilateral C3 channel exhibits ERS in the α (8–12 Hz) and β (13–30 Hz) bands, while the C4 channel in the contralateral area and the Cz channel in the central area demonstrate ERD in the α band. This ERS/ERD also occurs under the right-hand condition: pronounced ERD in the α band of the C3 channel and ERS in the α and β bands of the ipsilateral C4 channel.

By the time–frequency spectrograms, it could be confirmed that the generated MI EEG data own the subject's biological

information as well, indicating that the process of generating data in the SGD model realizes the capture and construction of the key characteristics of the MI brain activity. Also, this would help to solve the small sample problem of the MI BCI and further increase the decoding accuracy.

C. Limitation and Future Work

This study mainly improves the EEG data augmentation model by noise input and architecture of the GAN. Although the EEG decoding accuracy is improved, there still exist some limitations to be modified in the future.

First, the input of the GAN could be improved. This study replaces random noise with the EEG background noise as the input to the GAN, indicating that the EEG information is useful for generative learning. The extraction of EEG background noise in this study is ICA that decomposes the raw EEG data into ICs, assigns different weights according to correlation and irrelevance to EEG activity, and reorganizes them. Since EEG activity-related ICs are based on empirical judgment, the extraction may leave out some important or inherent biological information. In the future, it is better to design rules based on the ERS/ERD characteristics or cognitive functions to guide how to extract clear EEG background noise. Furthermore, rest-state EEG is easily collected when the subject is in an idle or rest state without any additional experiment. Applying this type of data is also a way to replace the random noise for the GAN. However, the public dataset does not contain this data, so in the future, we will try to conduct the MI experiment to collect these data.

Second, the architecture of the SGD could be improved on the following aspects: GAN structure, hyperparameter optimization method, and model interpretability. On the one hand, the proposed SGD model only uses a CNN structure as a generator and discriminator. Given that MI EEG samples are multichannel, long-duration data structures containing both temporal and spatial features, future work would develop a GAN that combines recurrent neural networks (RNNs) with CNNs to capture the long-term dependencies in MI EEG data, such as Xie et al.'s [23], Du et al.'s [32], Chen et al.'s [33], and Zhang et al.'s [34] works, facilitating the fusion of local and global information of EEG. On the other hand, the inherent intersubject difference in MI EEG data needs the hyperparameters to be subject-specific. Both the SGD and EBN modules in this study involve manual adjustment of parameters, which requires the engagement of specialized personnel and is time-consuming. In the future, automatic hyperparameter optimization techniques, such as Bayesian optimization and genetic algorithms, will be introduced to automatically adjust the hyperparameters of the SGD model and to reduce the need of manual intervention. In addition, although the model demonstrates strong performance in generating MI EEG samples, its complex architecture, particularly with multiple layers of convolution, skip connections, and attention mechanisms, makes the generation process hard to interpret. Model interpretability is crucial for neuroscience applications, especially those involving medical diagnosis or rehabilitation. Therefore, future work should aim to improve

the interpretability of the SGD model by incorporating visualization techniques to display changes in critical features during the generation process, which helps the researchers and clinicians better understand the generation mechanism and the generated MI EEG data.

Third, ablation experiments have demonstrated that skip connection facilitates the authenticity of the generated EEG data. The skip connection allows features in the encoder to be passed directly to the corresponding layer of the decoder. This means that all features in the encoder input (including MI-related components and artifacts) may influence the decoder output through skip connections. Therefore, future work would investigate the addition of gating mechanisms or attention modules to skip connections so that the model automatically learns and focuses on important MI features and ignores or suppresses artifactual features.

VI. CONCLUSION

BCI is a promising technology in the rehabilitation, monitor, and smart HCI fields, but the small sample problem of EEG decoding is one of the limitations of the development. The rapid growth of generative learning provides a new way to solve this problem. This study proposes a skip connection GAN-based data augmentation model to generate EEG samples for the MI-BCI, to expand the amount of the samples used for training the EEG decoding model. The EEG background noise is first proposed in our study to replace the random noise as the input of the GAN, and the combination of skip connection and multihead attention is designed for raw EEG data generation. With the help of the expansion of MI EEG data generated by SGD, the four-class EEG decoding accuracy is improved, indicating that the EEG background noise and proposed architecture can generate more authentic-like data. GAN or generative learning is promising to be an effective method to make up the gap between the MI-BCI experiment limitations and machine learning requirements, which prompts the development of high-accuracy EEG decoding for the BCI.

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